Compound interest tables

1

a

	Balance + interest	Total balance	Interest earned
First year	–	$4400	$220
Second year	$4400 + $220	$4620	$231
Third year	$4620 + $231	$4851	$242.55
Fourth year	$4851 + $242.55	$5093.55	–

b $693.55

2

a

	Balance + interest	Total balance	Interest earned
First year	–	$3600	$252
Second year	$3600 + $252	$3852	$269.64
Third year	$3852 + $269.64	$4121.64	$288.51
Fourth year	$4121.64 + $288.51	$4410.15	–

b $810.15

3

a

	Balance + interest	Total balance	Interest earned
First year	–	$570	$51.30
Second year	$570 + $51.30	$621.30	$55.92
Third year	$621.30 + $55.92	$677.22	–

b $107.22

4

a

	Balance + interest	Total balance	Interest earned
First year	–	$6100	$366
Second year	$6100 + $366	$6466	$387.96
Third year	$6466 + $387.96	$6853.96	–

b $753.96

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5

a

Time Period n (years)	Value at beginning of time period	Value at end of time period	Interest earned in time period
1	\$1000	\$1020	\$20
2	\$1020	\$1040.40	\$20.40
3	\$1040.40	\$1061.21	\$20.81

b \$61.21

Compound interest

6

a True **b** False **c** True **d** True

7

a \$81 033.75 **b** \$11 033.75

8

a \$647.20 **b** \$8737.20 **c** \$698.98 **d** \$9436.18

e \$754.89 **f** \$10 191.07 **g** \$2101.07 **h** \$25.97%

i \$1941.60 **j** \$159.47

9

a \$5894.42 **b** \$894.42

10 \$39 222.92

11 \$49 909.03

12

a $A(t) = 10\,000(1.013)^t$ **b** \$1009.40

Other compound periods

13

a \$8912.51 **b** \$5609 **c** \$1997.90 **d** \$5400.97

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14 **a** $2567.12 **b** $559.56 **c** $1400.60 **d** $4096.44

15 **a** $7761.19 **b** $7714.56 **c** $7690.89 **d** $7674.97
e $7668.82 **f** $7667.23

16 **a**
i $2265.59 **ii** $2268.86 **iii** $2270.13

b Compounding monthly earns $3.27 more than compounding quarterly and compounding weekly earns $1.27 more than compounding monthly.

c More interest is earned with more frequent compounding, but the extra interest earned decreases as the compounding frequency increases.

17 Plan 2 is the best option. We can either find the future value of each investment or calculate the effective annual interest rate of each plan.
Plan 1 has a future value of $6625.24, Plan 2 has a future value of $7050.97, and Plan 3 has a future value of $6855.10.

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